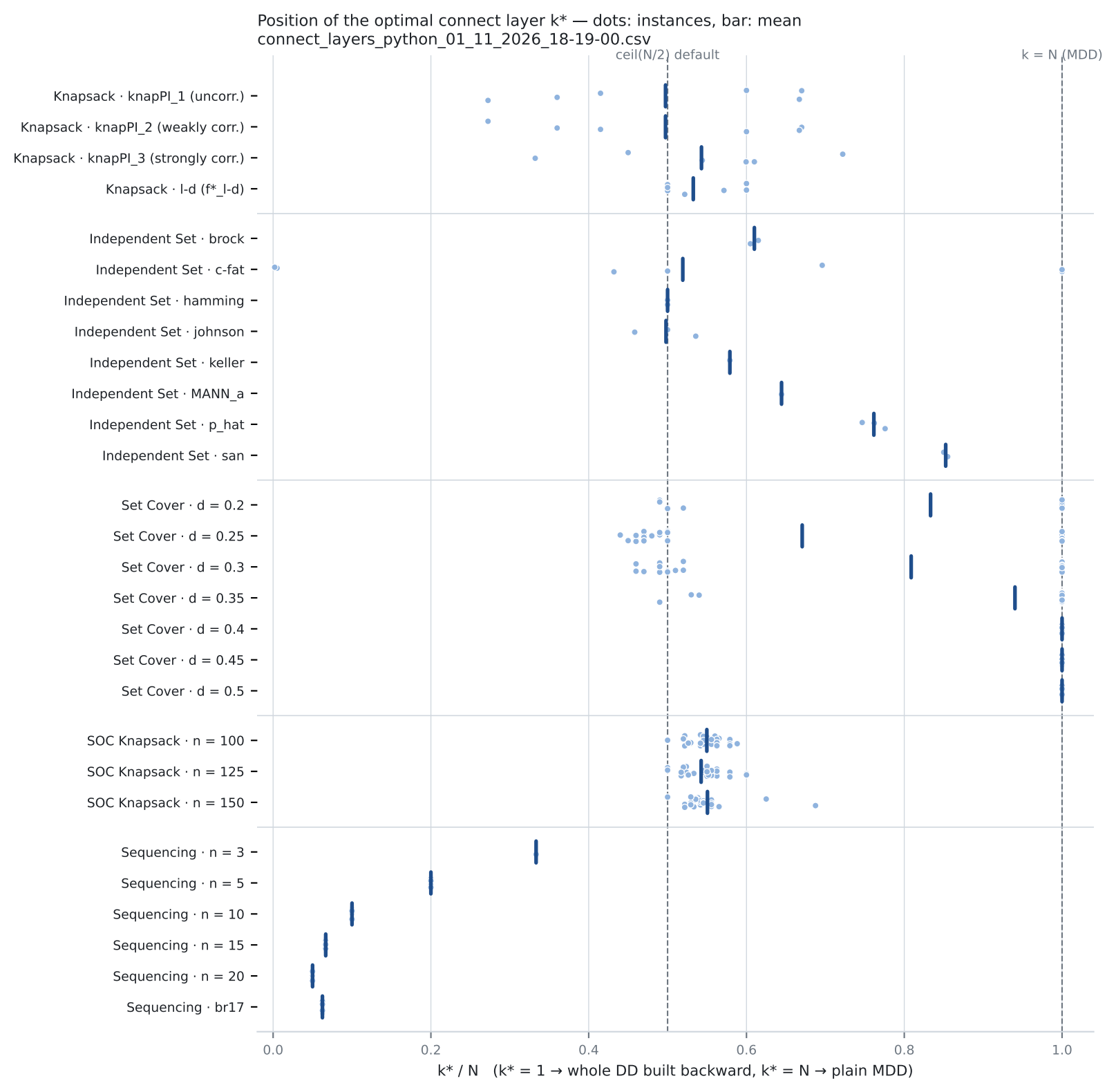

Position of the optimal connect layer k* — dots: instances, bar: mean
connect_layers_python_01_11_2026_18-19-00.csv
ceil(N/2) default
k = N (MDD)
Knapsack · knapPI_1 (uncorr.)
Knapsack · knapPI_2 (weakly corr.)
Knapsack · knapPI_3 (strongly corr.)
Knapsack · l-d (f*_l-d)
Independent Set · brock
Independent Set · c-fat
Independent Set · hamming
Independent Set · johnson
Independent Set · keller
Independent Set · MANN_a
Independent Set · p_hat
Independent Set · san
Set Cover · d = 0.2
Set Cover · d = 0.25
Set Cover · d = 0.3
Set Cover · d = 0.35
Set Cover · d = 0.4
Set Cover · d = 0.45
Set Cover · d = 0.5
SOC Knapsack · n = 100
SOC Knapsack · n = 125
SOC Knapsack · n = 150
Sequencing · n = 3
Sequencing · n = 5
Sequencing · n = 10
Sequencing · n = 15
Sequencing · n = 20
Sequencing · br17
0.0
0.2
0.4
0.6
0.8
1.0
k* / N (k* = 1 → whole DD built backward, k* = N → plain MDD)

Optimal connect layer k* per problem and instance family — DD Python
Source: connect_layers_python_01_11_2026_18-19-00.csv
k* minimizes the exact sMDD size before reduction (Σ_{j<k} forward + Σ_{j≥k} backward widths). N/2 = ceil(N/2), the default.
k=1 / <N/2 / =N/2 / >N/2 / k=N: number of instances in each position (shaded: the most common). 'nodes k*/N/2': geometric mean of nodes(k*) / nodes(ceil(N/2)).

Problem	Family	#	N	k*	k*/N mean	median	min–max	k=1	<N/2	=N/2	>N/2	k=N	nodes k*/N/2
Knapsack	knapPI_1 (uncorr.)	6	1466.7	625.0	0.50	0.51	0.27–0.67	0	3	0	3	0	0.99×
	knapPI_2 (weakly corr.)	6	1466.7	625.0	0.50	0.51	0.27–0.67	0	3	0	3	0	0.99×
	knapPI_3 (strongly corr.)	6	1466.7	667.5	0.54	0.57	0.33–0.72	0	2	0	4	0	1.00×
	l-d (f*_l-d)	9	11.4	6.0	0.53	0.50	0.50–0.60	0	0	8	1	0	0.99×
	All	**27**	**981.6**	**428.1**	**0.52**	**0.52**	**0.27–0.72**	**0**	**8**	**8**	**11**	**0**	**0.99×**
Independent Set	brock	2	200.0	122.0	0.61	0.61	0.60–0.61	0	0	0	2	0	0.69×
	c-fat	7	371.4	195.1	0.52	0.50	0.00–1.00	2	1	1	1	2	1.00×
	hamming	2	64.0	32.0	0.50	0.50	0.50–0.50	0	0	2	0	0	1.00×
	johnson	3	72.7	35.0	0.50	0.50	0.46–0.54	0	1	1	1	0	0.99×
	keller	1	171.0	99.0	0.58	0.58	0.58–0.58	0	0	0	1	0	0.80×
	MANN_a	1	45.0	29.0	0.64	0.64	0.64–0.64	0	0	0	1	0	1.00×
	p_hat	3	500.0	382.7	0.76	0.76	0.75–0.78	0	0	0	3	0	0.35×
	san	2	300.0	256.0	0.85	0.85	0.85–0.85	0	0	0	2	0	0.39×
	All	**21**	**269.6**	**169.9**	**0.60**	**0.60**	**0.00–1.00**	**2**	**2**	**4**	**11**	**2**	**0.75×**
Set Cover	d = 0.2	12	100.0	83.3	0.83	1.00	0.49–1.00	0	2	1	1	8	0.99×
	d = 0.25	32	100.0	67.1	0.67	0.49	0.44–1.00	0	17	3	0	12	0.96×
	d = 0.3	32	100.0	80.9	0.81	1.00	0.46–1.00	0	7	2	3	20	1.00×
	d = 0.35	32	100.0	94.0	0.94	1.00	0.49–1.00	0	1	0	3	28	0.99×
	d = 0.4	32	100.0	100.0	1.00	1.00	1.00–1.00	0	0	0	0	32	0.98×
	d = 0.45	32	100.0	100.0	1.00	1.00	1.00–1.00	0	0	0	0	32	0.98×
	d = 0.5	32	100.0	100.0	1.00	1.00	1.00–1.00	0	0	0	0	32	0.98×
	All	**204**	**100.0**	**89.9**	**0.90**	**1.00**	**0.44–1.00**	**0**	**27**	**6**	**7**	**164**	**0.98×**
SOC Knapsack	n = 100	30	20.3	11.1	0.55	0.55	0.50–0.59	0	0	8	22	0	0.91×
	n = 125	30	22.1	11.9	0.54	0.55	0.50–0.60	0	0	13	17	0	0.92×
	n = 150	30	23.9	13.0	0.55	0.54	0.50–0.69	0	0	3	27	0	0.87×
	All	**90**	**22.1**	**12.0**	**0.55**	**0.55**	**0.50–0.69**	**0**	**0**	**24**	**66**	**0**	**0.90×**
Sequencing	n = 3	1	3.0	1.0	0.33	0.33	0.33–0.33	1	0	0	0	0	1.00×
	n = 5	15	5.0	1.0	0.20	0.20	0.20–0.20	15	0	0	0	0	0.76×
	n = 10	15	10.0	1.0	0.10	0.10	0.10–0.10	15	0	0	0	0	0.53×
	n = 15	15	15.0	1.0	0.07	0.07	0.07–0.07	15	0	0	0	0	0.29×
	n = 20	15	20.0	1.0	0.05	0.05	0.05–0.05	15	0	0	0	0	0.26×
	br17	5	16.0	1.0	0.06	0.06	0.06–0.06	5	0	0	0	0	0.33×
	All	**66**	**12.6**	**1.0**	**0.10**	**0.07**	**0.05–0.33**	**66**	**0**	**0**	**0**	**0**	**0.42×**